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Configuration profiles

Abstract

In some examples, an electronic device includes an image sensor and a processor. The processor is to receive an image from the image sensor, identify multiple elements depicted in the image, and determine, based on the multiple elements identified, an environment. The processor is to retrieve a configuration profile associated with the environment and configure, according to a setting of the configuration profile, a peripheral device, an application, or a combination thereof.

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Background/Summary

BACKGROUND

(1) Electronic devices such as desktops, laptops, notebooks, tablets, smartphones, and mobile devices include peripheral devices, applications (e.g., machine-readable instructions that enable users to perform tasks), or a combination thereof, having configurable settings. The peripheral devices include audio devices (e.g., microphones, headsets, speakers), image sensors, display devices, privacy screens, network connections, or a combination thereof. The settings adjust all of or a portion of preferences associated with the peripheral devices, applications, or a combination thereof. The preferences include access to the peripheral devices, applications, or a combination thereof, as well as other parameters that affect operations of the peripheral devices, the applications, or a combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various examples are described below referring to the following figures.

(2) FIGS. **1**A-**1**D are images depicting environments of an electronic device that includes configuration profiles, in accordance with various examples.

(3) FIG. **2** is a flow diagram depicting elements of environments of an electronic device that

includes configuration profiles, in accordance with various examples.

(4) FIG. **3** is a block diagram depicting an electronic device for adjusting settings based on configuration profiles, in accordance with various examples.

(5) FIG. **4** is a block diagram depicting an electronic device for adjusting settings based on configuration profiles, in accordance with various examples.

(6) FIG. **5** is a block diagram depicting an electronic device for adjusting settings based on configuration profiles, in accordance with various examples.

DETAILED DESCRIPTION

(7) As described above, electronic devices include peripheral devices, applications, or a combination thereof, having settings that adjust access to the peripheral devices, the applications, or a combination thereof, as well as other parameters that affect operations of the peripheral devices, the applications, or a combination thereof. A peripheral device, as used herein, is a hardware device that enables sending or receiving of data from an electronic device to which the hardware device is coupled. The peripheral device may be internal or external to the electronic device. An application, as used herein, is machine-readable instructions that enable users to perform tasks (e.g., videoconferencing, live streaming, word processing, video streaming). The settings are adjustable based on an environment of the electronic device, for instance. The parameters include a volume level, a visibility level, availability of a feature, or a combination thereof, for instance.

(8) The environment, as used herein, refers to a physical environment in which the electronic device is located. Due to security reasons associated with the environment, availability of peripheral devices, or a combination thereof, a user adjusts the settings of the peripheral devices, the applications, or a combination thereof, for instance. The settings enable multiple configurations. For instance, in an office environment, an internal image sensor and an internal microphone are enabled while in a home environment, an external image sensor, an external speaker, and a headset are enabled. The settings may be adjusted at the system level through the operating system (OS) (e.g., machine-readable instructions that manage hardware components and applications of the electronic device) or at an application level. At the system level, the settings are applied regardless of which application is utilized. At the application level, the appearance and location of icons (e.g., graphical user interface (GUI) elements generated by machine-readable instructions) to adjust settings may vary. Having various methods with differing locations of icons for adjusting settings is confusing for the user and diminishes the user experience, and in instances where the user engages with an audience (e.g., videoconferencing, video calling, live streaming), diminishes the audience experience.

(9) To automate adjusting settings, an electronic device enables different configuration profiles of multiple configuration profiles. A configuration profile of the multiple configuration profiles includes settings for the peripheral devices, the applications, or a combination thereof, of the electronic device and an environment associated with the settings. The electronic device receives an image from an image sensor and analyzes the image to identify elements depicted in the image. The electronic device utilizes the elements identified in the image to determine an environment within which the electronic device is located. The electronic device enables a configuration profile of the multiple configuration profiles associated with the environment. Enabling the configuration profile associated with the environment adjusts settings of peripheral devices, applications, or a combination thereof, according to the settings of the configuration profile. For example, a first configuration profile is associated with an office environment and has a first setting that enables an internal image sensor and a second setting that enables an internal microphone. Responsive to the electronic device determining that elements depicted in an image indicate the office environment, the electronic device enables the internal image sensor and the internal microphone. A second configuration profile is associated with a home environment and has a first setting that enables an external image sensor and a second setting that enables a headset. Responsive to the electronic

device determining that elements depicted in an image indicate the home environment, the electronic device enables the external image sensor and the headset. In some examples, the electronic device receives an audio signal from a microphone and analyzes the audio file to determine the environment. In various examples, a user adjusts the settings, the environment, or a combination thereof, associated with a configuration profile. The user utilizes a graphical user interface (GUI) to adjust the settings, the environment, or a combination thereof, for example.

(10) By automatically adjusting settings according to the configuration profiles, the electronic device enhances the user and the audience experience because the electronic device is able to adjust settings of peripheral devices, applications, or a combination thereof, based on the environment of the electronic device. In some examples, the configuration profiles enable enhanced security by enabling privacy settings of peripheral devices, applications, or a combination thereof, or by disabling access to peripheral devices, applications, or a combination thereof, responsive to the environment of the electronic device.

(11) In some examples in accordance with the present description, an electronic device is provided. The electronic device includes an image sensor and a processor. The processor receives an image from the image sensor, identifies multiple elements depicted in the image, and determines, based on the multiple elements identified, an environment. The processor retrieves a configuration profile associated with the environment and configures, according to a setting of the configuration profile, a peripheral device, an application, or a combination thereof.

(12) In other examples in accordance with the present description, an electronic device is provided. The electronic device includes an image sensor, a storage device storing a first and a second configuration profile, and a processor. The first configuration profile includes a first setting and is associated with a first environment and the second configuration profile includes a second setting and is associated with a second environment. The processor receives an image from the image sensor and identifies an element depicted in the image. In response to the first environment including the element and the second environment excluding the element, the processor configures a peripheral device, an application, or a combination thereof, according to the first setting. In response to the second environment including the element and the first environment excluding the element, the processor configures the peripheral device, the application, or a combination thereof, according to the second setting.

(13) In yet other examples in accordance with the present description, a non-transitory machine-readable medium is provided. The term “non-transitory” does not encompass transitory propagating signals. The non-transitory machine-readable medium stores machine-readable instructions which, when executed by a processor of an electronic device, cause the processor to receive an image from an image sensor, identify multiple elements depicted in the image, determine multiple environments include the multiple elements, and cause a display device to display a GUI. The GUI includes a first configuration profile associated with a first environment of the multiple environments and a second configuration profile associated with a second environment of the multiple environments. The machine-readable instructions, when executed by the processor, cause the processor to, responsive to a selection of the first configuration profile, configure a peripheral device, an application, or a combination thereof, according to a first setting of the first configuration profile, and responsive to a selection of the second configuration profile, configure the peripheral device, the application, or a combination thereof, according to a second setting of the second configuration profile.

(14) Referring now to FIGS. **1**A-**1**D, images **100**, **108**, **120**, **141** of environments of an electronic device (e.g., the electronic device **300**, **400**, **500** described below with respect to FIGS. **3**, **4**, **5**, respectively) that includes configuration profiles are depicted, in accordance with various examples. The images **100**, **108**, **120**, **141** are captured by an image sensor of the electronic device, for example. FIG. **1**A includes an image **100**. The image **100** depicts an element **102**, **104**, **106**. The element **102** is a user, the element **104** is clothing, and the element **106** is a wall, for example. FIG.

1B includes an image **108**. The image **108** depicts an element **110**, **112**, **114**, **116**, **118**. The element **110** is a user, the element **112** is clothing, the element **114** is a wall, the element **116** is a clock, and the element **118** is a window, for example. FIG. **1**C includes an image **120**. The image **120** depicts an element **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**. The element **122** is a user, the element **124** is clothing, the element **126** is a wall, the element **128** is a table, the element **130** is a chair, the element **132** is a shelf, the elements **134**, **138** are sets of books, the element **136** is a picture, and the element **140** is a speaker. FIG. **1**D includes an image **141**. The image **141** depicts an element **142**, **144**, **146**, **148**, **150**, **152**, **154**. The element **142** is a user, the element **144** is clothing, the element **146** is a headset, the element **148** is a wall, the element **150** is a window, the element **152** is a clock, and the element **154** is a chair.

(15) In some examples, the user represented by the element **102** is the user represented by the element **110**, **122**, **142**. In other examples, the user represented by the element **102** is the user represented by the element **110** and is a different user than the user represented by the element **122**, **142**. For example, the electronic device is a shared electronic device and the user represented by the element **102** is a first user, the user represented by the element **122** is a second user, and the user represented by the element **142** is a third user. In some examples, the clock represented by the element **116** is the clock represented by the element **122**, and the wall represented by the element **114** is the wall represented by the element **148**. For example, the image **108** captures a first environment that is a first location of a room and the image **141** captures a second environment that is a second location of the room.

(16) As described above, the electronic device receives the images **100**, **108**, **120**, **141** from the image sensor and analyzes the images **100**, **108**, **120**, **141** to identify elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154**, respectively, depicted in the images **100**, **108**, **120**, **141**. In some examples, the image sensor captures the image responsive to a user launching an application (e.g., videoconferencing, live streaming, video calling). In other examples, the image sensor captures the image responsive to a modification of a network connection. In yet other examples, the image sensor captures the image responsive to the electronic device powering on or waking from a sleep state. The electronic device identifies the elements **102**, **104**, **106**, **110**, **112**, **114**, **116**, **118**, **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154** utilizing image processing techniques, machine learning techniques, or a combination thereof, for example. In some examples, the electronic device utilizes image processing techniques to analyze the images **100**, **108**, **120**, **141** to identify the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154**, respectively. For example, the electronic device decomposes the images **100**, **108**, **120**, **141** to identify the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154**, respectively. The image processing techniques include grayscaling, blurring, thresholding, dilating, erosion, or a combination thereof. For example, the electronic device converts the images **100**, **108**, **120**, **141** to a grayscale image. The grayscale image has color removed to enhance the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154** of the images **100**, **108**, **120**, **141**, respectively. The electronic device blurs the grayscale image to remove noise. The electronic device thresholds the blurred image to convert the blurred image into pixels that are either black or white.

(17) In various examples, the electronic device determines that the white pixels indicate the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154** of the images **100**, **108**, **120**, **141**, respectively. In various examples, responsive to the decomposed image including multiple elements (e.g., the elements **102**, **104**, **106** of the image **100**; the elements **110**, **112**, **114**, **116**, **118** of the image **108**; the elements **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140** of the image **120**; the elements **142**, **144**, **146**, **148**, **150**, **152**, **154** of the image **141**), the electronic device determines that a first subset of the

multiple elements represents a first element and a second subset of the multiple elements represents a second element by identifying connected subsets of the multiple elements. A connected subset of the multiple elements, as used herein, is a group of pixels of the decomposed image having a same value (e.g., a same color) and that touch in a contiguous manner to form an outline. In various examples, the electronic device dilates (e.g., add pixels to), erodes (e.g., remove pixels from), or a combination thereof, the thresholded image to enhance the element **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154** of the image **100**, **108**, **120**, **141**, respectively, for identification.

(18) In some examples, the electronic device utilizes a machine learning technique such as a convolutional neural network (CNN) to analyze the images **100**, **108**, **120**, **141** to identify the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154**, respectively. For example, the electronic device utilizes a region-based CNN (R-CNN). The electronic device divides the images **100**, **108**, **120**, **141** into multiple regions. In various examples, the electronic device utilizes a Region Proposal Network (RPN) to determine the multiple regions. The electronic device inputs the multiple regions into the R-CNN to detect the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154**, respectively.

(19) In various examples, the electronic device utilizes a computer vision technique to identify the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**,**126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154** of the images **100**, **108**, **120**, **141**, respectively. The computer vision techniques utilize machine learning networks (e.g., CNN, R-CNN) trained with data sets including a wide variety of elements located within a wide variety of environments. The computer vision techniques include object detection, semantic segmentation, instance segmentation, or a combination thereof. For example, the electronic device utilizes panoptic segmentation, which is a combination of semantic and instance segmentation. Utilizing semantic segmentation, the electronic device categorizes the elements **102**, **104**, **106**; **110**, **112**, **114**, **116**, **118**; **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; **142**, **144**, **146**, **148**, **150**, **152**, **154** into classes. Utilizing instance segmentation, the electronic device identifies individual elements within the classes. For example, utilizing panoptic segmentation, the electronic device analyzes the image **120** and identifies the elements **134**, **138** as the sets of books. The electronic device determines that the elements **134**, **138** include multiple elements and that each element of the multiple elements is an individual book. In various examples, the sets of books indicate a class and the individual elements indicate a label associated with the class. For example, the individual book title is a label associated with the set of books.

(20) Referring now to FIG. **2**, a flow diagram **200** for an electronic device (e.g., the electronic device **300**, **400**, **500** described below with respect to FIGS. **3**, **4**, **5**, respectively) that includes configuration profiles is depicted, in accordance with various examples. The flow diagram **200** includes elements **210**, **212**, **214**, **216**, **218**, **220**, **222**, **224**, **226**, **228**, **230**, **232** (e.g., the elements **102**, **104**, **106**, **110**, **112**, **114**, **116**, **118**, **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**) and environments **202**, **204**, **206**, **208**. The elements **210**, **212**, **214**, **216**, **218**, **220**, **222**, **224**, **226**, **228**, **230**, **232** are herein collectively referred to as the elements **210-232**. The flow diagram **200** is a graphical representation that shows the relationships between the elements **210-232** and the environments **202**, **204**, **206**, **208**. The environment **202** indicates a first environment L1 (e.g., an environment captured by the image **141**). The environment **204** indicates a second environment L2 (e.g., an environment captured by the image **120**). The environment **206** indicates a third environment L3 (e.g., an environment captured by the image **108**). The environment **208** indicates a fourth environment L4 (e.g., an environment captured by the image **100**). The environment **202** is associated with the elements **214**, **216**, **218**, **222**, **224**. The environment **204** is associated with the elements **218**, **224**, **226**, **228**, **230**, **232**. The environment **206** is associated with the elements **210**, **212**, **222**, **228**. The environment **208** is associated with the

elements **218**, **220**, **226**.
(21) The flow diagram **200** is a graphical representation of a data structure (e.g., linked list, database, hash table, arrays, heaps, graphs, trees) storing the relationships between the elements **210-232** and the environments **202**, **204**, **206**, **208**, for example. In some examples, the data structure is stored to a storage device of the electronic device, as shown and described below with respect to FIG. **4**. In other examples, the data structure is stored to a remote storage device communicatively coupled to the electronic device, as shown and described below with respect to FIG. **3**. The remote storage device is communicatively coupled to the electronic device via a network connection, for example.
(22) The elements **210-232** include classes, labels, identifiers, audio files or characteristics of audio files, or a combination thereof. The characteristics of an audio file include an amplitude, a frequency, a background noise, a bandwidth, a power level, or any other measurable characteristic. In some examples, the elements **210-232** include an identifier of a user or identifiers of multiple users, an identifier of an electronic device or identifiers of multiple electronic devices, an identifier of a network connection or identifiers of multiple network connections, an identifier of an application or identifiers of multiple applications, an identifier of a peripheral device or identifiers of multiple peripheral devices, or a combination thereof. In various examples, the identifier of a user of the multiple users, the identifier of an electronic device of the multiple electronic devices, the identifier of a network connection of the multiple network connections, the identifier of an application of the multiple applications, the identifier of a peripheral device of the multiple peripheral devices, or a combination thereof, is associated with multiple configuration profiles.
(23) As described above, the electronic device utilizes the elements **210-232** identified from the image (e.g., the image **100**, **108**, **120**, **141**) to determine an environment **202**, **204**, **206**, **208** within which the electronic device is located. For example, utilizing the techniques described above with respect to FIG. **1**, the electronic device analyzes the image and identifies the elements **214**, **216**, **218**, **222**, **224**. The electronic device determines that the elements **214**, **216**, **218**, **222**, **224** are associated with the environment **202** and that the electronic device is located in the environment **202**. In another example, utilizing the techniques described above with respect to FIG. **1**, the electronic device analyzes the image and identifies the elements **218**, **224**. The electronic device determines that the elements **218**, **224** are associated with the environments **202**, **204**. In some examples, as described above, the electronic device receives an audio signal from a microphone and analyzes the audio file to determine whether the electronic device is located in the environment **202** or the environment **204**. For example, the electronic device analyzes frequency differences in different portions of the audio signal, patterns of energy concentration across the different portions of the audio signal, differences between energy concentration patterns of the different portions of the audio signal, or a combination thereof. In some examples, based on the analysis of the audio signal, the electronic device determines that the audio signal indicates the user is located in an environment having no background noise. The electronic device determines whether other elements of the environments **202**, **204** are elements that indicate no background noise. For example, responsive to the electronic device determining that the element **216** associated with the environment **202** indicates no background noise and that none of the elements **226**, **228**, **230**, **232** associated with the environment **204** indicate no background noise, the electronic device determines that the electronic device is located in the environment **202**.
(24) In other examples, to determine whether the electronic device is located in the environment **202** or the environment **204**, the electronic device determines whether the elements **218**, **224** indicate classes, as described above with respect to FIG. **1**. Responsive to a determination that the element **218** indicates a class, the electronic device determines whether the image includes a label associated with the class, for example. In some examples, the flow diagram **200** includes additional elements (not explicitly shown) that are associated with an element of the elements **210-232** and an environment of the environments **202**, **204**, **206**, **208**. In other examples, the flow diagram **200**

includes a second layer of elements (not explicitly shown) that include labels associated with elements that indicate classes and an environment of the environments **202**, **204**, **206**, **208**. The electronic device determines whether the image depicts an element that is associated with a label. For example, the element **218** indicates clothing (e.g., the elements **104**, **112**, **124**, **144**). The electronic device determines that the image includes clothing that is associated with a "tie" label (e.g., the element **112**). Responsive to a determination that the image includes the element that is associated with the label, the electronic device determines the environment (e.g., the environment of the image **108**) based on the environment associated with the label.

(25) In various examples, the environments **202**, **204**, **206**, **208** correspond to a first configuration profile of multiple configuration profiles, a second configuration profile of multiple configuration profiles, a third configuration profile of multiple configuration profiles, and a fourth configuration profile of multiple configuration profiles, respectively. The first configuration profile includes a first group of settings, the second configuration profile includes a second group of settings, the third configuration profile includes a third group of settings, and a fourth configuration profile includes a fourth group of settings. A group of settings, as used herein, may include an individual setting or multiple settings. As described above, the settings of the configuration profile indicate that the electronic device is to adjust all or a portion of preferences associated with the peripheral devices, applications, or a combination thereof. The preferences include access to the peripheral devices, applications, or a combination thereof, as well as other parameters that effect operations of the peripheral devices, the applications, or a combination thereof. For example, the settings indicate whether to mute or unmute of a microphone, whether to enable or disable the microphone, whether to enable or disable an external speaker, a level of brightness of a display device, a volume level of the microphone, a volume level of the headset, a volume level of a speaker, whether to enable or disable an application or a portion of a functionality of the application, whether to enable or disable a privacy screen, whether to enable or disable a network connection, or a combination thereof. In another example, a setting is associated with an application and indicates that the electronic device is to cause a display device to display a GUI of an application, where the GUI includes a subset of multiple options, the subset indicated by the setting. In yet another example, a setting is associated with a text-to-speech application and indicates that electronic device is to cause the text-to-speech application to filter out words or phrases not associated with the environment.

(26) Referring now to FIG. **3**, a block diagram of an electronic device **300** for adjusting settings based on configuration profiles is depicted, in accordance with various examples. The electronic device **300** is a desktop, a laptop, a notebook, a tablet, a smartphone, or any other suitable computing device that interacts with peripheral devices, applications, or a combination thereof, for example. The electronic device **300** includes a processor **302**, an image sensor **304**, and a storage device **306**. The processor **302** is a microprocessor, a microcomputer, a microcontroller, a programmable integrated circuit, a programmable gate array, or another suitable processor or controller for managing operations of the electronic device **300**. The processor **302** is a central processing unit (CPU), a graphics processing unit (GPU), a system on a chip (SoC), an image signal processor (ISP), or a field programmable gate array (FPGA), for example. The image sensor **304** is an internal camera, an external camera, or any other suitable device for capturing an image (e.g., the image **100**, **108**, **120**, **141**), recording a video signal, or a combination thereof. The storage device **306** is a hard drive, a solid state drive (SSD), flash memory, random access memory (RAM), or other suitable memory for storing data or machine-readable instructions of the electronic device **300**.

(27) While not explicitly shown, the electronic device **300** includes network interfaces, video adapters, sound cards, local buses, peripheral devices (e.g., a keyboard, a mouse, a touchpad, a speaker, a microphone, a display device, a privacy screen), wireless transceivers, connectors, or a combination thereof. While the image sensor **304** is shown as an integrated image sensor of the electronic device **300**, in other examples, the image sensor **304** couples to any suitable connection

for enabling communications between the electronic device **300** and the image sensor **304**. The connection may be via a wired connection (e.g., a Universal Serial Bus (USB)) or via a wireless connection (e.g., BLUETOOTH®, WI-FI®). In some examples, the processor **302**, the image sensor **304**, the storage device **306**, or a combination thereof, are located within an integrated circuit such as an embedded artificial intelligence (eAI) chip.

(28) In some examples, the processor **302** couples to the image sensor **304** and the storage device **306**. The storage device **306** stores machine-readable instructions, which, when executed, cause the processor **302** to perform some or all of the actions attributed herein to the processor **302**. The machine-readable instructions are the machine-readable instructions **308**, **310**, **312**, **314**, **316**.

(29) In various examples, the machine-readable instructions **308**, **310**, **312**, **314**, **316** cause the processor **302** to adjust settings based on configuration profiles. The machine-readable instruction **308**, when executed by the processor **302**, causes the processor **302** to receive the image from the image sensor **304**. The machine-readable instruction **310**, when executed by the processor **302**, causes the processor **302** to identify multiple elements (e.g., the elements **102**, **104**, **106**; the elements **110**, **112**, **114**, **116**, **118**; the elements **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; the elements **142**, **144**, **146**, **148**, **150**, **152**, **154**) depicted in the image. The processor **302** identifies the multiple elements depicted in the image utilizing the techniques described above with respect to FIG. **1**, for example. The machine-readable instruction **312**, when executed by the processor **302**, causes the processor **302** to determine, based on the multiple elements identified, an environment (e.g., the environment **202**, **204**, **206**, **208**). The processor **302** determines the environment utilizing the techniques described above with respect to FIG. **2**, for example. The machine-readable instruction **314**, when executed by the processor **302**, causes the processor **302** to retrieve a configuration profile associated with the environment. The machine-readable instruction **316**, when executed by the processor **302**, causes the processor **302** to configure, according to a setting of the configuration profile, a peripheral device, an application, or a combination thereof.

(30) In some examples, the configuration profile is stored on a storage device of the electronic device **300**, as shown and described below with respect to FIG. **4**. In other examples, a remote storage device (not explicitly shown) stores the configuration profile. The remote storage device is a remotely managed storage device such as an enterprise cloud, a public cloud, a data center, a server farm, or some other suitable remotely managed storage device, for example. The electronic device accesses the remote storage device via a network connection (not explicitly shown), for example. To retrieve the configuration profile associated with the environment, the processor **302** causes a transmission that includes the environment determined by executing the machine-readable instruction **312** to the remote storage device. The processor **302** receives a transmission that includes the configuration profile associated with the environment from the remote storage device.

(31) As described above, the electronic device **300** enables different configuration profiles of multiple configuration profiles, where a configuration profile of the multiple configuration profiles includes settings for the peripheral devices, the applications, or a combination thereof, of the electronic device **300**. In various examples, responsive to receipt of the transmission from the remote storage device, the processor **302** configures, according to a setting the configuration profile associated with the environment, a peripheral device, an application, or a combination thereof. For example, the image sensor **304** is the peripheral device, and the setting of the configuration profile associated with the environment, indicates that the processor **302** is to disable the image sensor **304**. In various examples, the configuration profile associated with the environment includes another setting for another peripheral device, another application, or a combination thereof. For example, another setting of the configuration profile associated with the environment indicates that the processor **302** is to enable an external image sensor (not explicitly shown).

(32) As described above, the environment associated with the configuration profile is associated with an element that is an identifier of a user. In some examples, the processor **302** determines the identifier of the user utilizing the electronic device **300**. For example, the processor **302** utilizes

facial recognition techniques to identify a user of the electronic device **300** and determines the identifier of the user based on the identification. The processor **302** retrieves the configuration profile associated with the environment and the identifier of the user utilizing the electronic device **300**. The processor **302** configures, according to a setting of the configuration profile associated with the environment and the identifier of the user utilizing the electronic device **300**, a peripheral device, an application, or a combination thereof. For example, a living room environment is associated with a first user and a second user. The first user utilizes a first volume for a speaker, and the second user utilizes a second volume for the speaker. In some examples, the second volume is different than the first volume. The processor **302** determines that the identifier of the first user is utilized to access the electronic device **300**. The processor **302** retrieves the configuration profile associated with the environment and the identifier of the first user. The processor **302** configures the speaker according to the setting of the configuration profile that is associated with the identifier of the first user.

(33) As described above, the environment associated with the configuration profile is associated with an element that is an identifier of an electronic device. In some examples, the processor **302** determines the identifier of the electronic device **300**. The processor **302** retrieves the configuration profile associated with the environment and the identifier of the electronic device **300**. The processor **302** configures, according to a setting of the configuration profile associated with the environment and the identifier of the electronic device **300**, a peripheral device, an application, or a combination thereof. For example, a public environment (e.g., a coffee shop, a library) is associated with a first electronic device and a second electronic device. The first electronic device utilizes a headset and an internal image sensor, and the second electronic device utilizes the headset and disables the internal image sensor. The processor **302** determines that the identifier of the electronic device **300** indicates the second electronic device. The processor **302** retrieves the configuration profile associated with the public environment and the second electronic device. The processor **302** configures the internal image sensor according to the setting of the configuration profile that is associated with the environment and the second electronic device.

(34) Referring now to FIG. **4**, a block diagram depicting an electronic device **400** for adjusting settings based on configuration profiles **416** is depicted, in accordance with various examples. The electronic device **400** is the electronic device **300**, for example. The electronic device **400** includes a processor **402**, an image sensor **404**, and a storage device **406**. The processor **402** is the processor **302**, for example. The image sensor **404** is the image sensor **304**, for example. The storage device **406** is the storage device **306**, for example.

(35) In some examples, the processor **402** couples to the image sensor **404** and the storage device **406**. The storage device **406** stores machine-readable instructions, which, when executed, cause the processor **402** to perform some or all of the actions attributed herein to the processor **402**. The machine-readable instructions are the machine-readable instructions **408**, **410**, **412**, **414**. The storage device **406** stores the configuration profiles **416**. The configuration profiles **416** include a configuration profile **418**, **420**. The configuration profile **418** includes a setting **422** and is associated with an environment **424**. The configuration profile **420** includes a setting **426** and is associated with an environment **428**.

(36) In various examples, the machine-readable instructions **408**, **410**, **412**, **414** cause the processor **402** to adjust settings based on the configuration profiles **416**. The machine-readable instruction **408**, when executed by the processor **402**, causes the processor **402** to receive an image (e.g., the image **100**, **108**, **120**, **141**) from the image sensor **404**. The machine-readable instruction **410**, when executed by the processor **402**, causes the processor **402** to identify an element (e.g., the element **102**, **104**, **106**, **110**, **112**, **114**, **116**, **118**, **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**, **142**, **144**, **146**, **148**, **150**, **152**, **154**) depicted in the image. To identify the element, the processor **402** utilizes the techniques described above with respect to FIG. **2**, for example. In response to the first environment including the element and the second environment excluding the element, the

machine-readable instruction **412**, when executed by the processor **402**, causes the processor **402** to configure a peripheral device, an application, or a combination thereof, according to the first setting. The processor **402** determines whether the element is associated with the first environment, the second environment, or a combination thereof, utilizing the techniques described above with respect to FIG. **2**, for example. In response to the second environment including the element and the first environment excluding the element, the machine-readable instruction **414**, when executed by the processor **402**, causes the processor **402** to configure the peripheral device, the application, or a combination thereof, according to the second setting.

(37) In some examples, the processor **402**, in response to the first environment and the second environment excluding the element, stores a third configuration profile to the storage device **406**. The third configuration profile includes a third setting to configure the peripheral device, the application, or a combination thereof, and is associated with a third environment that includes the element. In other examples, in response to the first environment and the second environment including the element, the processor **402** identifies another element depicted in the image. In response to the first environment including the another element and the second environment excluding the another element, the processor **402** configures a peripheral device, an application, or a combination thereof, according to the first setting. In response to the second environment including the another element and the first environment excluding the another element, the processor **402** configures the peripheral device, the application, or a combination thereof, according to the second setting. In response to the first environment and the second environment excluding the element and the another element, the processor **402** stores a third configuration profile to the storage device **406**. The third configuration profile includes a third setting to configure the peripheral device, the application, or a combination thereof, and is associated with a third environment that includes the element and the another element.

(38) In various examples, in response to the first environment and the second environment including the element, the processor **402** receives an audio file via a microphone (not explicitly shown). The processor **402** determines the audio file indicates the first environment. The processor **402** determines the audio file indicates the first environment utilizing the techniques described above with respect to FIG. **2**, for example. The processor **402** configures the peripheral device, the application, or a combination thereof, according to the first setting of the first configuration profile.

(39) In other examples, in response to an inability to identify an element depicted in the image, the processor **402** receives the audio file via the microphone (not explicitly shown). The processor **402** determines that the audio file indicates the first environment utilizing the techniques described above with respect to FIG. **2**, for example. The processor **402** configures the peripheral device, the application, or a combination thereof, according to the first setting of the first configuration profile. In another example, the processor **402** determines that the audio file does not indicate the first or the second environments. In response to the determination that the audio file does not indicate the first or the second environments, the processor **402** stores a third configuration profile to the storage device **406**. The third configuration profile includes a third setting to configure the peripheral device, the application, or a combination thereof, and is associated with a third environment that includes the element and the audio file.

(40) Referring now to FIG. **5**, a block diagram depicting an electronic device **500** for adjusting settings based on configuration profiles (e.g., the configuration profiles **416**) is depicted, in accordance with various examples. The electronic device **500** is the electronic device **300**, **400**, for example. The electronic device **500** includes a processor **502** and a non-transitory machine-readable medium **504**. The processor **502** is the processor **302**, **402**, for example. The non-transitory machine-readable medium **504** is the storage device **306**, **406**, for example. The non-transitory machine-readable medium **504** stores machine-readable instructions, which, when executed, cause the processor **502** to perform some or all of the actions attributed herein to the processor **502**. The machine-readable instructions are the machine-readable instructions **506**, **508**,

510, **512**, **514**, **516**.

(41) In various examples, the machine-readable instructions **506**, **508**, **510**, **512**, **514**, **516** cause the processor **502** to adjust settings based on the configuration profiles. The machine-readable instruction **506**, when executed by the processor **502**, causes the processor **502** to receive an image (e.g., the image **100**, **108**, **120**, **141**) from an image sensor (e.g., the image sensor **304**, **404**). The machine-readable instruction **508**, when executed by the processor **502**, causes the processor **502** to identify multiple elements (e.g., the elements **102**, **104**, **106**; the elements **110**, **112**, **114**, **116**, **118**; the elements **122**, **124**, **126**, **128**, **130**, **132**, **134**, **136**, **138**, **140**; the elements **142**, **144**, **146**, **148**, **150**, **152**, **154**) depicted in the image. The machine-readable instruction **510**, when executed by the processor **502**, causes the processor **502** to determine that multiple environments (e.g., the environments **202**, **204**, **206**, **208**) include the multiple elements (e.g., the elements **210-232**). The processor **502** determines the multiple environments include the multiple elements utilizing the techniques described above with respect to FIG. **2**, for example. The machine-readable instruction **512**, when executed by the processor **502**, causes the processor **502** to cause a display device (not explicitly shown) to display a GUI. The GUI includes a first configuration profile (e.g., the configuration profile **418**) associated with a first environment (e.g., the environment **424**) of the multiple environments and a second configuration profile (e.g., the configuration profile **420**) associated with a second environment (e.g., the environment **428**) of the multiple environments. The machine-readable instruction **514**, when executed by the processor **502**, causes the processor **502** to, responsive to a selection of the first configuration profile, configure a peripheral device, an application, or a combination thereof, according to a first setting (e.g., the setting **422**) of the first configuration profile. The machine-readable instruction **516**, when executed by the processor **502**, causes the processor **502** to, responsive to a selection of the second configuration profile, configure the peripheral device, the application, or a combination thereof, according to a second setting (e.g., the setting **426**) of the second configuration profile.

(42) In various examples, the processor **502** determines that the multiple elements are excluded from the multiple environments. In response to the determination that the multiple elements are excluded from the multiple environments, the processor **502** causes the display device to display the GUI to prompt a user to verify that the multiple elements are associated with a third environment. In response to a verification that the multiple elements are associated with the third environment, the processor **502** generates a third configuration profile, the third configuration profile associated with the third environment and including a third setting for the peripheral device, the application, or a combination thereof. In some examples, the processor **502** stores the third configuration profile to the non-transitory machine-readable medium **504**. In other examples, the processor **502** causes transmission of the third configuration profile to a remote storage device that stores the configuration profiles.

(43) In some examples, the processor **502** receives another image from the image sensor. The processor **502** identifies another set of multiple elements depicted in the another image. The processor **502** determines that the third environment includes the another set of multiple elements. In response to the determination that the third environment includes the another set of multiple elements, the processor **502** configures the peripheral device, the application, or a combination thereof, according to the third setting of the third configuration profile.

(44) In other examples, the processor **502** determines that the multiple elements are excluded from the multiple environments. The processor **502** determines a class of an element of the multiple elements. The processor **502** causes the display device to display the GUI prompting a user to verify a third environment associated with a third configuration profile, the third environment including the class of the element. In response to a verification of the third environment, the processor **502** configures the peripheral device, the application, or a combination thereof, according to a third setting for the peripheral device, the application, or a combination thereof.

(45) In some examples, in response to a determination that multiple environments include the

multiple elements, the processor **502** receives an audio file via a microphone (not explicitly shown). The processor **502** determines that the audio file indicates the first environment. In response to the determination that the audio file indicates the first environment, the processor **502** configures the peripheral device, the application, or a combination thereof, according to the first setting of the first configuration profile.

(46) In various examples, in response to a determination that the multiple elements are excluded from the multiple environments, the processor **502** determines degrees of matching of the multiple elements with the multiple environments. For example, the processor **502** determines that a first environment of the multiple environments include 80% of the multiple elements, a second environment of the multiple environments include 92% of the multiple elements, a third environment of the multiple environments include 10% of the multiple elements, and a fourth environment of the multiple environments include 25% of the multiple environments. In some examples, the processor **502** causes the GUI to display the first configuration profile associated with the first environment and the second configuration profile associated with the second environment.

(47) In other examples, in response to a determination that the multiple elements are excluded from the multiple environments, the processor **502** eliminates a first subset of the multiple elements that are common to the multiple environments. The processor **502** determines degrees of matching of a second subset of the multiple elements with the multiple environments, where the second subset includes the remaining elements of the multiple elements. For example, the processor **502** determines that a first environment of the multiple environments include 10% of the second subset of multiple elements, a second environment of the multiple environments include 50% of the multiple elements, a third environment of the multiple environments include 10% of the multiple elements, and a fourth environment of the multiple environments include 25% of the multiple environments. In some examples, the processor **502** causes the GUI to display the second configuration profile associated with the second environment and the fourth configuration profile associated with the fourth environment.

(48) In some examples, in response to a determination that a peripheral device, an application, or a combination thereof, associated with the first setting of the first configuration profile, the processor **502** causes the GUI to display text that informs the user that the peripheral device, the application, or the combination thereof, is unavailable and prompts the user to couple the peripheral device to the electronic device **500**, to select another configuration profile, or a combination thereof. In other examples, the processor **502** configures another peripheral device, another application, or a combination thereof to a default setting. For example, responsive to the first setting indicating that the processor **502** is to enable a headset and a determination that the headset is unavailable, the processor **502** is to enable an internal microphone and an internal speaker.

(49) By detecting elements and determining a degree of matching with environments of the configuration profiles, the electronic device can automatically adjust settings based on the degree of matching. By automatically adjusting the settings according to a configuration profile associated with an environment that has a highest degree of matching with the detected elements, the electronic device enhances the user and the audience experience because the electronic device is able to adjust settings of peripheral devices, applications, or a combination thereof, based on the similarity of the current environment of the electronic device with the environment associated with the configuration profile.

(50) As described herein, the terms “application,” “software,” and “firmware” are considered to be interchangeable in the context of the examples provided. “Firmware” is considered to be machine-readable instructions that a processor of the electronic device executes prior to execution of the operating system (OS) of the electronic device, with a small portion that continues after the OS bootloader executes (e.g., a callback procedure). “Application” and “software” are considered broader terms than “firmware,” and refer to machine-readable instructions that execute after the OS

bootloader starts, through OS runtime, and until the electronic device shuts down. “Application,” “software,” and “firmware,” as used herein, are referred to as executable code.
(51) The above description is meant to be illustrative of the principles and various examples of the present description. Numerous variations and modifications become apparent to those skilled in the art once the above description is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.
(52) In the figures, certain features and components disclosed herein are shown in exaggerated scale or in somewhat schematic form, and some details of certain elements are not shown in the interest of clarity and conciseness. In some of the figures, in order to improve clarity and conciseness, a component or an aspect of a component is omitted.
(53) In the above description and in the claims, the term “comprising” is used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . . .” Also, the term “couple” or “couples” is intended to be broad enough to encompass both direct and indirect connections. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices, components, and connections. Additionally, the word “or” is used in an inclusive manner. For example, “A or B” means any of the following: “A” alone, “B” alone, or both “A” and “B.”

Claims

1. An electronic device, comprising: an image sensor; and a processor to: receive an image from the image sensor; identify multiple elements depicted in the image; determine, based on the multiple elements identified, an environment; retrieve a configuration profile associated with the environment; and configure, according to a setting of the configuration profile, a peripheral device, an application, or a combination thereof.
2. The electronic device of claim 1, wherein a remote storage device is to store the configuration profile.
3. The electronic device of claim 1, wherein the image sensor is the peripheral device.
4. The electronic device of claim 1, wherein the processor is to identify the multiple elements utilizing panoptic segmentation.
5. The electronic device of claim 1, wherein the configuration profile includes another setting for another peripheral device, another application, or a combination thereof.
6. An electronic device, comprising: an image sensor; a storage device storing a first and a second configuration profile, the first configuration profile including a first setting and associated with a first environment and the second configuration profile including a second setting and associated with a second environment; and a processor to: receive an image from the image sensor; identify an element depicted in the image; in response to the first environment including the element and the second environment excluding the element, configure a peripheral device, an application, or a combination thereof, according to the first setting; and in response to the second environment including the element and the first environment excluding the element, configure the peripheral device, the application, or a combination thereof, according to the second setting.
7. The electronic device of claim 6, wherein the processor is to, in response to the first environment and the second environment excluding the element, store a third configuration profile to the storage device, the third configuration profile including a third setting and associated with a third environment, the third setting to configure the peripheral device, the application, or a combination thereof, the third environment including the element.
8. The electronic device of claim 6, wherein, in response to the first environment and the second environment including the element, the processor is to: identify another element depicted in the image; in response to the first environment including the another element and the second environment excluding the another element, configure a peripheral device, an application, or a

combination thereof, according to the first setting; and in response to the second environment including the another element and the first environment excluding the another element, configure the peripheral device, the application, or a combination thereof, according to the second setting.
9. The electronic device of claim 8, wherein, in response to the first environment and the second environment excluding the element and the another element, the processor is to store a third configuration profile to the storage device, the third configuration profile including a third setting and associated with a third environment, the third setting to configure the peripheral device, the application, or a combination thereof, the third environment including the element and the another element.
10. The electronic device of claim 6, wherein, in response to the first environment and the second environment including the element, the processor is to: receive an audio file via a microphone; determine that the audio file indicates the first environment; and configure the peripheral device, the application, or a combination thereof, according to the first setting of the first configuration profile.
11. A non-transitory machine-readable medium storing machine-readable instructions which, when executed by a processor of an electronic device, cause the processor to: receive an image from an image sensor; identify multiple elements depicted in the image; determine that multiple environments include the multiple elements; in response to the determination that the multiple environments include the multiple elements, cause a display device to display a graphical user interface (GUI), the GUI including a first configuration profile associated with a first environment of the multiple environments and a second configuration profile associated with a second environment of the multiple environments; responsive to a selection of the first configuration profile, configure a peripheral device, an application, or a combination thereof, according to a first setting of the first configuration profile; and responsive to a selection of the second configuration profile, configure the peripheral device, the application, or a combination thereof, according to a second setting of the second configuration profile.
12. The non-transitory machine-readable medium of claim 11, wherein, in response to a determination that the multiple elements are excluded from the multiple environments, the processor is to: cause the display device to display the GUI to prompt a user to verify that the multiple elements are associated with a third environment; and in response to a verification that the multiple elements are associated with the third environment, generate a third configuration profile, the third configuration profile associated with the third environment and including a third setting for the peripheral device, the application, or a combination thereof.
13. The non-transitory machine-readable medium of claim 12, wherein the processor is to: receive another image from the image sensor; identify another set of multiple elements depicted in the another image; determine that the third environment includes the another set of multiple elements; and in response to the determination that the third environment includes the another set of multiple elements, configure the peripheral device, the application, or a combination thereof, according to the third setting of the third configuration profile.
14. The non-transitory machine-readable medium of claim 11, wherein, in response to a determination that the multiple elements are excluded from the multiple environments, the processor is to: determine a class of an element of the multiple elements; cause the display device to display the GUI prompting a user to verify a third environment associated with a third configuration profile, the third environment including the class of the element; and in response to a verification of the third environment, configure the peripheral device, the application, or a combination thereof, according to a third setting for the peripheral device, the application, or a combination thereof.
15. The non-transitory machine-readable medium of claim 11, wherein, in response to a determination that the multiple environments include the multiple elements, the processor is to: receive an audio file via a microphone; determine that the audio file indicates the first environment;

and configure the peripheral device, the application, or a combination thereof, according to the first setting of the first configuration profile.